



Finite time convergence of Kuramoto model

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Introduction

Consider the potential function $V(\theta) = \frac{1}{2} \sum_{1 \leq i, j \leq N} \frac{K}{N} (1 - \cos(\theta_i - \theta_j))$. It is known that for the differential equation

$$\dot{\theta}_i = \omega_i - \frac{\partial V(\theta)}{\partial \theta_i} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin \theta_j - \theta_i, \forall i = 1, 2, \dots, N$$

The solution of the differential equation will synchronize asymptotically with large enough coupling strength K , i.e.

$$\lim_{t \rightarrow \infty} |\dot{\theta}_i(t) - \dot{\theta}_j(t)| = 0, \forall i, j = 1, 2, \dots, N$$

In this project, we introduce the method of achieving finite time convergence of this kind of Kuramoto model and we have further improved the convergent time of the model.

1 Numerical experiment of signed gradient model

The model in vector form Dong & Xue¹ use to achieve finite time convergence is

$$\dot{\theta} = \omega - \text{sign}(\nabla V(\theta))$$

where we assume all the oscillators are identical, i.e., $\omega_1 = \omega_2 = \dots = \omega_N$. Also, the differential equation is subject to the initial condition, $\theta(0) = \theta^0$.

We can discover the convergent time of this kind of model is finite through several numerical experiments. Note that $D(\theta^0) = \max_{i \in \{1, \dots, N\}}(\theta_i^0) - \min_{i \in \{1, \dots, N\}}(\theta_i^0)$

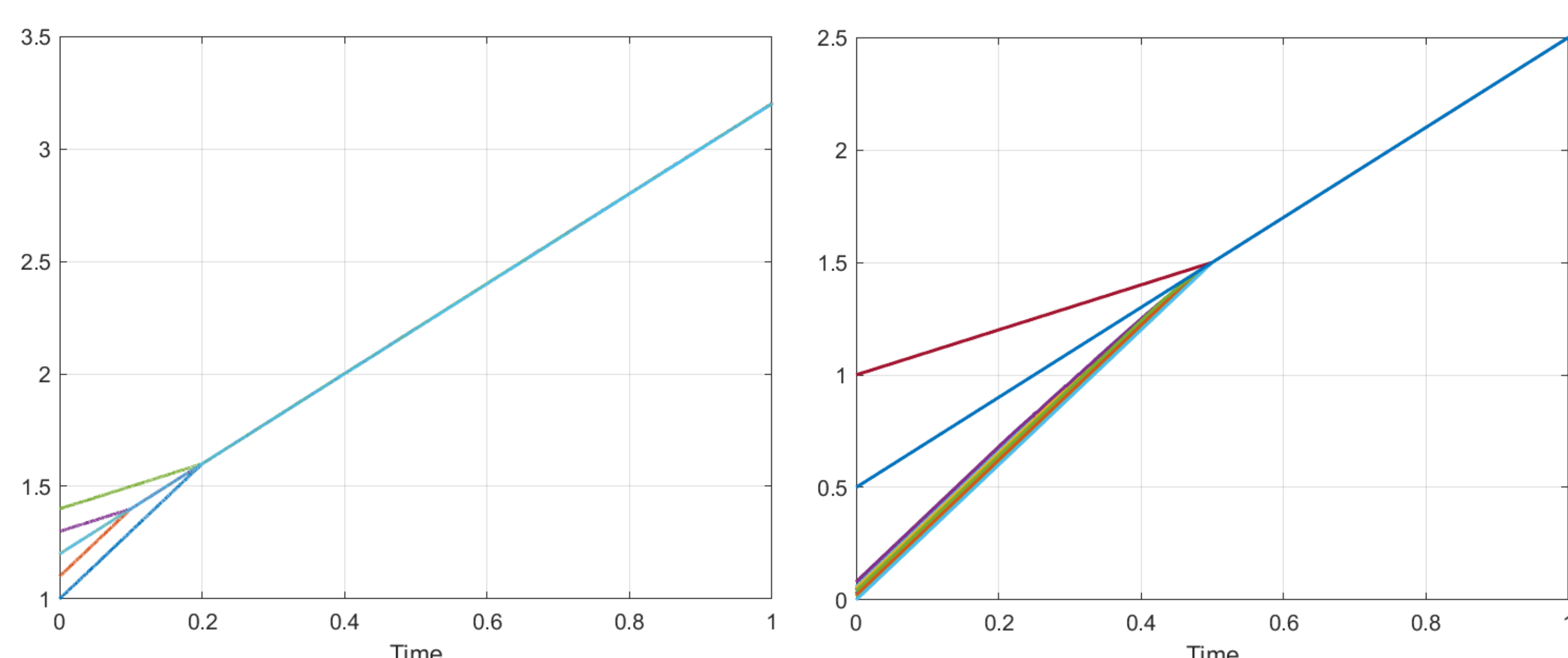


Figure 1: The example from left to right are with the initial configuration of $N=4$ and $\theta^0 = [1.0, 1.1, 1.2, 1.3, 1.4]$ with estimated convergent time $T=0.2$, and $N=14$ and $\theta^0 = [0.0284, 0.0436, 0.0489, 0.0588, 0.0156, 0.0584, 0.0805, 0.0454, 0.0226, 0.0518, 0.0793, 0.0400, 0, 1]$ with estimated convergent time $T=0.5$

$D(\theta^0)$	N	Convergent time	$D(\theta^0)$	N	Convergent time
0.1	14	0.049501	1.0	3	0.49981
0.2	14	0.099504	1.0	4	0.49962
0.3	14	0.1495	1.0	5	0.49966
0.4	14	0.1995	1.0	6	0.49953
0.5	14	0.24955	1.0	7	0.49953
0.6	14	0.29956	1.0	8	0.49953

Table 1: The convergent time of signed gradient model with respect to different initial configuration θ^0 and N .

From the experiments, we can observe that the estimated convergent time is roughly equal to $\frac{D(\theta^0)}{2}$. We will show this result in the next section.

2 Finite time convergence

For differential equation with discontinuous right hand side, we understand the solution often in Filippov sense. That is, the solution in the interval $[t_1, t_2]$ of the equation

$$\dot{\theta} = \omega - \text{sign}(\nabla V(\theta)) = X(\theta)$$

is defined as the solution $\theta(t) : [t_1, t_2] \rightarrow \mathbf{R}^N$ satisfying absolute continuous and $\dot{\theta}(t) \in K[X](t)$ almost all $t \in [t_1, t_2]$.

$$K[X](t) = \bigcap_{\delta > 0} \bigcap_{\mu(S)=0} \text{co}\{X(B(t, \delta) \setminus S)\}$$

where $\mu(S) = 0$ is Lebesgue measure zero.

Let $\Omega_f \subset \mathbf{R}^N$ be the set where f cannot be differentiable and S be any set of measure zero. The generalized gradient of f at $x \in \mathbf{R}^N$ is defined by

$$\partial f = \text{co}\{\lim_{i \rightarrow \infty} df(x_i) | x_i \rightarrow x, x_i \notin S \cup \Omega_f\}$$

Thus, the directional derivative of f along with a vector field X at θ can be defined as

$$\tilde{\mathcal{L}}_X(\theta) = \{a \in \mathbf{R} | \exists v \in K[X](\theta) \text{ with } \zeta \cdot v = a, \forall \zeta \in \partial f\}$$

The following result is taken from Cortes²

Theorem 2.1 (LaSalle Invariance Principle). *Let $f : \mathbf{R}^d \rightarrow \mathbf{R}$ be a locally Lipschitz and regular function. Let $x_0 \in S \subset \mathbf{R}^d$, with S compact and strongly invariant. Assume either $\max \tilde{\mathcal{L}}_X f(x) \leq 0$ or $\tilde{\mathcal{L}}_X f(x) = \emptyset$ for all $x \in S$. Let $Z_{X,f} = \{x \in \mathbf{R}^d | 0 \in \tilde{\mathcal{L}}_X f(x)\}$. Then, any solution $x : [t_0, +\infty) \rightarrow \mathbf{R}^d$ starting from x_0 converges to the largest weakly invariant set M contained in $\overline{Z_{X,f}} \cap S$. Moreover, if the set M is a finite collection of points, then the limit of all solutions starting at x_0 exists and equals one of them.*

Theorem 2.2 (Finite-time convergence with first-order information). *Under the same assumptions of Theorem 2.1, further assume that there exists a neighborhood U of $Z_{X,f} \cap S$ in S such that $\max \tilde{\mathcal{L}}_X f \leq -\epsilon < 0$ a.e. on $U \setminus (Z_{X,f} \cap S)$. Then, any solution $x : [x_0, +\infty) \rightarrow \mathbf{R}^d$ starting from S reaches $Z_{X,f} \cap S$ in finite time. Moreover, if $U=S$, then the convergence time is upper bounded by $(f(x_0) - \min_{x \in S} f(x))/\epsilon$*

Corollary 2.2.1. *If $\tilde{\mathcal{L}}_X f = -\epsilon < 0$ a.e. on $U \setminus (Z_{X,f} \cap S)$ and $U=S$, then the convergence time is $(f(x_0) - \min_{x \in S} f(x))/\epsilon$*

We can then apply corollary 2.2.1 to find the convergent time.

Let $f(\theta) = \frac{\max_{i \in \{1, \dots, N\}}(\theta_i) - \min_{i \in \{1, \dots, N\}}(\theta_i)}{2}$, $U = S = \mathbf{R}^N$, and $Z_{X,f} = \{\theta = (\theta_1, \theta_2, \dots, \theta_N) | \theta_1 = \theta_2 = \dots = \theta_N\}$. $\forall a \in \tilde{\mathcal{L}}_X(\theta)$. Then, there exists a $v \in K[X](\theta)$ such that for all $\zeta \in \partial f$, we have $v \cdot \zeta = a$. Take θ with $f(\theta) \neq 0$. Let $e_M = \max_{i \in \{1, 2, \dots, N\}}(\theta_i)$ and $e_m = \min_{i \in \{1, 2, \dots, N\}}(\theta_i)$. Then, $\frac{1}{2}(e_M - e_m) \in \partial f(\theta)$. Since $v_M = \omega - 1$ and $v_m = \omega + 1$, $v \cdot \zeta = -1$.

And from Corollary 2.2.1, we have the convergence time, which is

$$f(\theta^0) = \frac{\max_{i \in \{1, \dots, N\}}(\theta_i) - \min_{i \in \{1, \dots, N\}}(\theta_i)}{2} = \frac{D(\theta^0)}{2}$$

3 Conclusion

- The convergence time have been improved from the upper bound of $\frac{|\nabla V(\theta^0)|_1}{\lambda_0}$ where $\lambda_0 = \min_{\theta \in H} \lambda_2(\text{Hess}(V)(\theta))$ and $H = \{\theta | \max_{1 \leq i \leq N} \theta_i + \min_{1 \leq i \leq N} \theta_i, D(\theta) \leq D(\theta^0)\}$ to $\frac{D(\theta^0)}{2}$

References

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